ONE OUTPUT · 1 / 10

One row meets one column.

Build one output, then look for inputs it can share.

> K is an index. The highlighted cells hold input values.

FOUR BY FOUR MATRICES · OUTPUT C[0,0]

A · row 0

	0	1	2	3
0	−2.0	−1.0	0.0	1.0
1	−1.0	0.0	1.0	2.0
2	0.0	1.0	2.0	−2.0
3	1.0	2.0	−2.0	−1.0

B · column 0

	0	1	2	3
0	−1.0	0.0	1.0	−1.0
1	0.0	1.0	−1.0	0.0
2	1.0	−1.0	0.0	1.0
3	−1.0	0.0	1.0	−1.0

Match A[0,k] with B[k,0] for each K position.

C[0,0] = ?

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ONE OUTPUT · 2 / 10

Add four products.

Other outputs can need the same input values.

> One thread per output is a chosen teaching mapping.

FOUR BY FOUR MATRICES · OUTPUT C[0,0]

A · row 0

	0	1	2	3
0	−2.0	−1.0	0.0	1.0
1	−1.0	0.0	1.0	2.0
2	0.0	1.0	2.0	−2.0
3	1.0	2.0	−2.0	−1.0

B · column 0

	0	1	2	3
0	−1.0	0.0	1.0	−1.0
1	0.0	1.0	−1.0	0.0
2	1.0	−1.0	0.0	1.0
3	−1.0	0.0	1.0	−1.0

K = 0	K = 1	K = 2	K = 3
−2.0 × −1.0	−1.0 × 0.0	0.0 × 1.0	1.0 × −1.0
2.0	**0.0**	**0.0**	**−1.0**

2.0 + 0.0 + 0.0 + (−1.0) = 1.0

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SHARED INPUT TILES · 3 / 10

Load two small input tiles.

Eight loaded values feed four partial outputs.

> These are the first output tile's partial sums, not the complete C.

FIRST PHASE · K POSITIONS 0–1

A tile

Rows 0–1 · K 0–1

r0, k0	r0, k1
−2.0	−1.0
r1, k0	r1, k1
−1.0	0.0

B tile

K 0–1 · columns 0–1

k0, c0	k0, c1
−1.0	0.0
k1, c0	k1, c1
0.0	1.0

Partial C

After K 0–1

r0, c0	r0, c1
2.0	−1.0
r1, c0	r1, c1
1.0	0.0

4 A inputs + 4 B inputs = 8 values loaded

Four threads build four partial output sums.

Input tiles in shared memory · one private sum per thread

Teaching algorithm · caches and GPU timing excluded

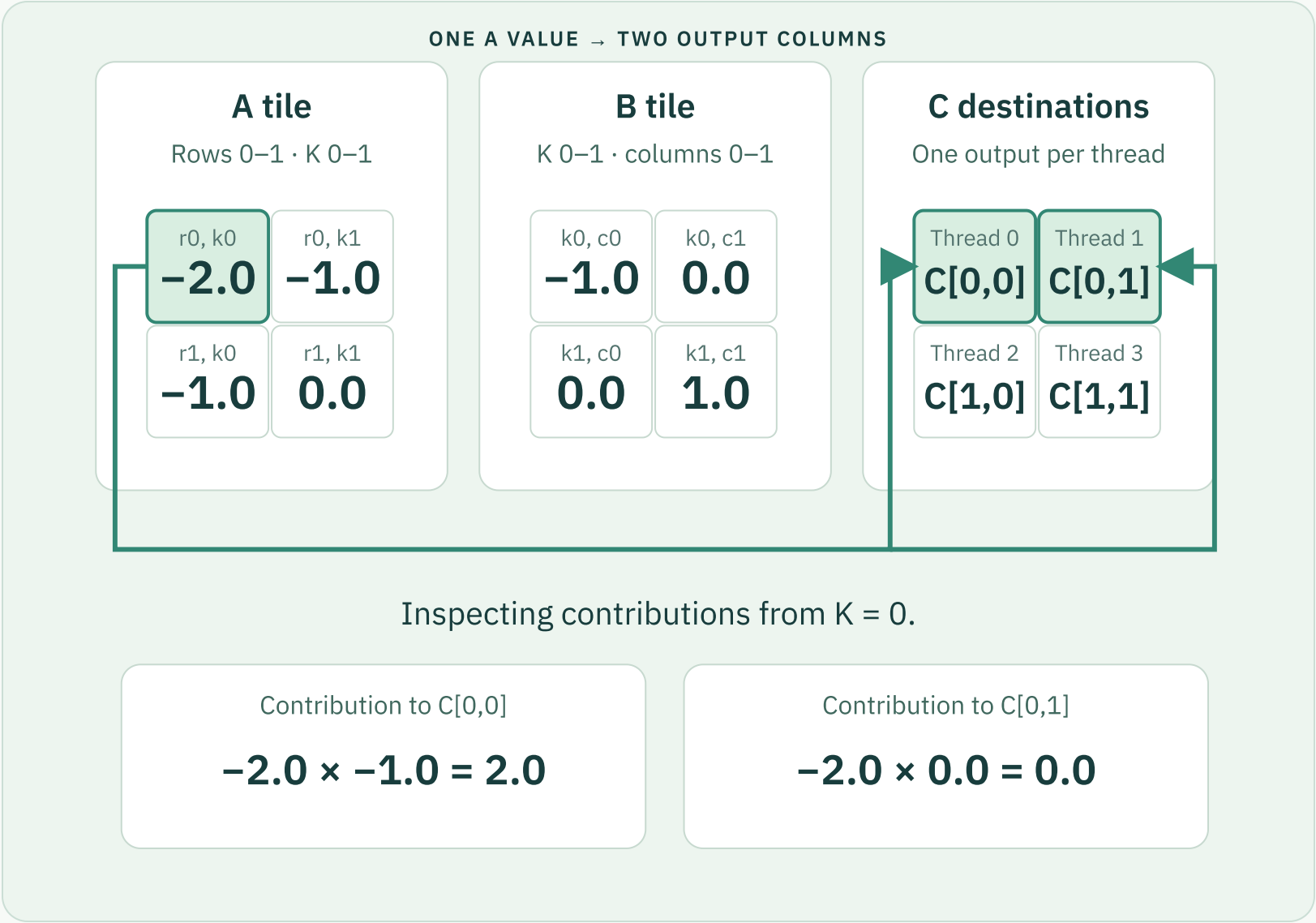
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SHARED INPUT TILES · 4 / 10

Reuse an A value across columns.

The same loaded value contributes to two outputs.

> Arrows identify the outputs using one selected input.



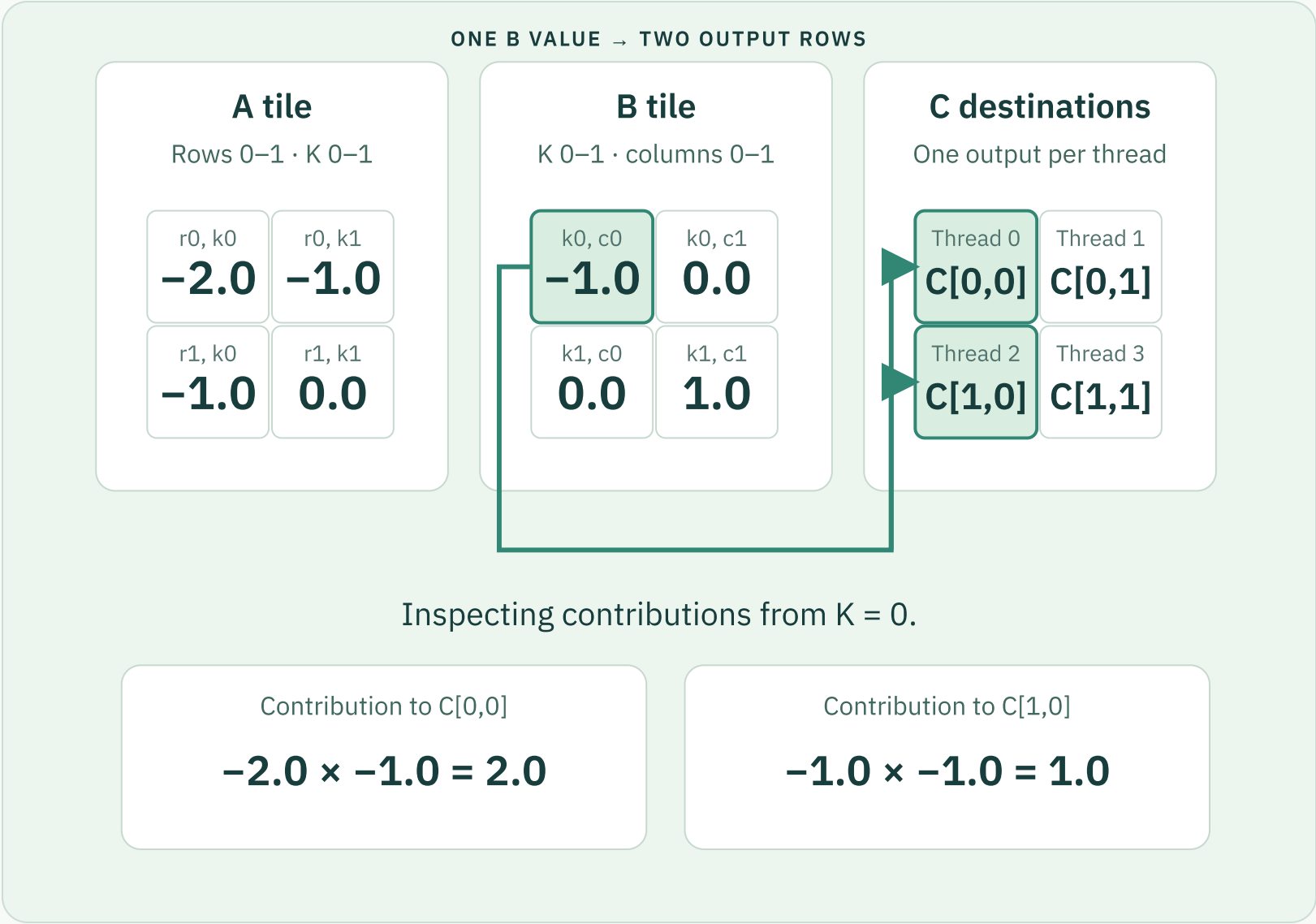
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SHARED INPUT TILES · 5 / 10

Reuse a B value across rows.

Sharing works in both input tiles.

> The same B input can serve different output rows.



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SHARED INPUT TILES · 6 / 10

Keep the sums. Replace the inputs.

Finish using the current tiles before overwriting them.

> Shared inputs change; each thread keeps its private partial sum.

NEXT INPUT TILES · PREVIOUS SUMS RETAINED

A tile

Rows 0–1 · K 2–3

r0, k2 **0.0**	r0, k3 **1.0**
r1, k2 **1.0**	r1, k3 **2.0**

B tile

K 2–3 · columns 0–1

k2, c0 **1.0**	k2, c1 **−1.0**
k3, c0 **−1.0**	k3, c1 **0.0**

Retained partial C

After K 0–1

r0, c0 **2.0**	r0, c1 **−1.0**
r1, c0 **1.0**	r1, c1 **0.0**

Finish old reads → replace tiles → order new writes

The threads retain their own partial sums.

A new input phase adds to them; it does not reset them.

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SHARED INPUT TILES · 7 / 10

Add the next phase's contribution.

The partial sums carry across K phases.

> Input tiles may be reused only after their users finish.

SECOND PHASE · K POSITIONS 2–3

A tile

Rows 0–1 · K 2–3

r0, k2	r0, k3
0.0	1.0
r1, k2	r1, k3
1.0	2.0

B tile

K 2–3 · columns 0–1

k2, c0	k2, c1
1.0	−1.0
k3, c0	k3, c1
−1.0	0.0

Completed C tile

After K 0–3

r0, c0	r0, c1
1.0	−1.0
r1, c0	r1, c1
0.0	−1.0

C[0,0]: 2.0 + (−1.0) = 1.0

All four K positions have contributed.

The output matches the separate-output reference.

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COMPARE THE ALGORITHMS · 8 / 10

Same output. Fewer explicit loads.

Compare the complete four by four multiplication.

> Input-load counts cover all output tiles and all K phases.

COMPLETE FOUR BY FOUR MULTIPLICATION

Same C · 16 output values

	0	1	2	3
0	1.0	−1.0	0.0	1.0
1	0.0	−1.0	1.0	0.0
2	4.0	−1.0	−3.0	4.0
3	−2.0	4.0	−2.0	−2.0

Explicit input values loaded

Tile width 1 · 128 input loads



Tile width 2 · 64 input loads



Tile width 4 · 32 input loads



Caches excluded · counts are not elapsed time

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TRY A NEW CASE · 9 / 10

Pause: use a wider tile.

Eight by eight matrices. Tile width four.

> Pause before checking the next slide.

ONE OUTPUT TILE · ONE K PHASE · N = 8



How many input values load in this phase?

How many outputs can use each loaded input?

Pause and explain the scope of this sharing.

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TRY A NEW CASE · 10 / 10

Sharing belongs to an output tile.

Two input tiles, each with sixteen values.

> Return to Chapter 9 → compare tile widths while keeping inputs fixed.

ONE OUTPUT TILE · ONE K PHASE · N = 8



16 + 16 = 32 input values in this phase

Each input serves 4 outputs within this tile.

Another output tile can load the same input again.

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